

# OSCAR DE LA CERDA

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## EDUCATION

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**Yale University**, New Haven, CT

May 2028

- *ABET in Electrical Engineering*
- *Certificate in Programming*
- *Relevant Coursework:* Electromagnetism (PHYS 181), Introduction to Computer Engineering (EENG 201), Introduction to Communications and Control (ECE 202), Circuits and Systems Design (ECE 2031), Data Structures and Programming (CS 223), Digital Systems (ECE 348)

**University of Iceland**, Reykjavik, Iceland

June – July 2025

- *Relevant Coursework:* Renewable Energy Technology, Politics and Economics of Renewables

**Tom C. Clark High School**, San Antonio, Texas

June 2024

- *Awards:* Questbridge National College Match Finalist, 14 AP courses, National Hispanic Recognition from College Board, AP Scholar with Distinction from College Board

## TECHNICAL SKILLS

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- *Computer Languages:* C, C++, MATLAB, Racket, Verilog, PowerShell, Bash
- *Hardware and Electronics:* Circuit Design, Breadboard Electronics, Oscilloscopes, soldering, PCB assembly, Arduino, Sensor integration, Integrated circuitry, FPGA encoding
- *Tools:* Altium, LTSpice

## EXPERIENCE

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**Yale Undergraduate Aerospace Association**, *Sub team Lead*

November 2024 - Now

- Designed, simulated, and implemented an analog-to-digital converter (ADC) circuit to enable a Teensy microcontroller to process pressure, voltage, and current sensor data during pre-fuel testing
- Developed a complete avionics sensor-suite schematic integrating GPS, IMU, altimeter, MCU, and S-PLIG modules to improve system reliability and flight data acquisition
- Implemented SPI communication between the ADC and microcontroller for high-speed digital transfer of sensor measurements.

**Yale Bulldogs Racing**, *Sub team Member*

November 2024 – May 2025

- Assembled and soldered a high-voltage PCB for the team's Formula-style electric race car, supporting power-delivery reliability in competitive environments
- Authored technical documentation detailing internal electrical components to ensure compliance with safety and racing regulations

## PROJECTS & LEADERSHIP EXPERIENCE

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**Piezoelectricity in Iceland**, *Independent Project*

June – July 2025

- Modeled and evaluated piezoelectric energy-harvesting systems (cantilever beams, diaphragm configurations, harvesting eels) on Icelandic ships to offset greenhouse gas emissions
- Conducted modeling on MATLAB to quantify electrical energy that could be captured in North Atlantic Ocean waves

**Spanish National Honor Society**, *President*, San Antonio, Texas

2023 - 2024

- Led a 50-member community service organization representing diverse Latin American and European backgrounds, fostering social engagement and support for those in need

**National Honor Society**, *Vice - President*, San Antonio, Texas

2023 - 2024

- Led a community service-based organization of about 100 students by creating and coordinating 20 service opportunities among various levels: local, citywide, and global
- Partnered with UNICEF's Côte d'Ivoire project – Organized a partnership with UNICEF to create a fund that would help transform plastic waste along Ivory Coast, Africa into safe and sustainable building materials to help build classrooms that are urgently needed. Raised a total of \$2,000